CREO SYLLABUS		
OBJECTIVE: The objective of this training Session is to impart knolwedge for the students about Creo on Solid fundamentals of the CAD tool with Parts modeling, assembling and sheets. The training session is to prepare the student for more specific and advanced functions		
Duration		
S.No	**CONTENT**	**HOURS (L/P)**
1	Introduction to Creo Parametric	
	Feature-Based Nature	
	Bidirectional Associative Property	
	Parametric Nature	
	System Requirements	
	Getting Started with Creo Parametric	
	Important Terms and Definitions	
	File Menu Options	
	Managing Files Menu	
	Manager Model Tree	
	Understanding the Functions of the Mouse Buttons	
	Ribbon	
	Toolbars	
	Navigator	
	Creo Parametric Browser	
	Appearance Gallery	
	Rendering in Creo Parametric	
	Colour Scheme used in this book	
2	Creating Sketches in the Sketch Mode-I	
	The Sketch Mode	
	Working with the Sketch Mode	
	Invoking the Sketch Mode	
	The Sketcher Environment	
	Working with a Sketch in the Sketch Mode	
	Drawing a Sketch Using tools available in the Sketch Tab	
	Placing a Point	
	Drawing a Line	
	Drawing a Centreline	
	Drawing a Geometry Centreline	
	Drawing a Rectangle	
	Drawing a Circle	
	Drawing an Ellipse	
	Drawing an Arc	
	Dimensioning the Sketch	
	Converting a Weak Dimension into a Strong Dimension	
	Dimensioning a Sketch Using the Normal Tool	
	Dimensioning the Basic Sketched Entities	
	Linear Dimensioning of a Line	
	Angular Dimensioning of an Arc	
	Diameter Dimensioning Radial	
	Revolved Sections	
	Working with Constraints	
	Types of Constraints	

	Disabling Constraints	
	Modifying the Dimensions of a Sketch	
	Using the Modify Button	
	Modifying a Dimension by Double-Clicking on it	
	Modifying Dimensions Dynamically	
	Resolve Sketch Dialog Box	
	Deleting the Sketched Entities	
	Trimming the Sketched Entities	
	Mirroring the Sketched Entities	
	Inserting Standard/User-Defined Sketches	
	Drawing Display Options	
3	Creating Sketches in the Sketch Mode-II	
	Dimensioning the Sketch	
	Dimensioning a Sketch Using the Baseline Tool	
	Replacing the Dimensions of a Sketch Using the Replace Tool	
	Creating Fillets	
	Creating Circular Fillets	
	Creating Elliptical Fillets	
	Creating a Reference Coordinate System	
	Working with Splines	
	Creating a Spline	
	Dimensioning of Splines	
	Modifying a Spline	
	Writing Text in the Sketcher Environment	
	Rotating and Resizing Entities	
	Importing 2D Drawings in the Sketch Mode	
4	Creating Base Features	
	Creating Base Features	
	Invoking the part mode	
	The default datum planes	
	Creating a protrusion	
	Extruding a sketch	
	Revolving a sketch	
	Understanding the Orientation of datum planes	
	Parent Child Relationship	
	Implicit Relationship	
	Explicit Relationship	
	Nesting of Sketches	
5	Datums	
	Default Datum Planes Need for	
	Datums in modeling Selection	
	Method in Creo Parametric Datum	
	Options	
	Datum Planes	
	Creating Datum Planes	
	Datum Planes Created on-the-fly	
	Ddatum Axes	
	Datum Points	
	Creating Cuts	

	Removing Material by using the extrude Tool	
	Removing Material by using the revolve Tool	
6	Options Aiding Construction of Parts - 1	
	Creating Holes	
	The hole Dashboard	
	Important Points to remember while creating a hole	
	Creating Rounds	
	Creating Basic Rounds	
	Creating a Variable Radius Round	
	Points to remember while creating Rounds	
	Creating Chamfers	
	Corner Chamfer	
	Edge Chamfer	
	Understanding Ribs	
	Creating Trajectory Ribs	
	Creating Profie Ribs	
	Editing Features of a model	
	Editing Defenitiion or Redefining Features	
	Reordering Features	
	Rerouting Features	
	Suppresing Features	
	Deleting Features	
	Modifying Features	
7	Options Aiding Construction of Parts - 2	
	Introduction	
	Creating Featuer Patterns	
	Uses of patterns	
	Deleting s Pattern	
	Copying Features	
	New Refs	
	Same Refs	
	Mirroring the Sketched Entities	
	Move	
	Select	
	Mirroring a Geometry	
	Creating a Section of a Solid Model	
	Work Region Method	
8	Advanced Modeling Tools - I	
	Other Protrusion Options	
	Sweep Features	
	Creating Sweep Protrusions	
	Aligning a sketched Trajectory to an existing Geometry	
	Creating a thin Sweep Protrusion	
	Creating Sweep Cut	
	Blend Features	
	Parallel Blend	
	Rotational Blend General Blend	
	General Blend	
	Using Blend Vortex	

	Shell Feature	
	Creating a Constant Thickness Shell	
	Creating a variable Thickness Shell 8	
	Datum Curves	
	Creating a Datum Curve by using the curve Button	
	Creating a Datum Curve by sketching	
	Creating a Curve by using the Intersect Option	
	Creating a Curve by using the project Option	
	Creating a curve by using the wrap Option	
	Creating Draft Features	
9	Advanced Modeling Tools - II	
	Advanced Feature Creation Tools	
	Variable Section Sweep Using the Sweep Option	
	Swept Blend	
	Helical Sweep	
	Blend Section to Surfaces	
	Blend Between Surfaces	
10	Advanced Modeling Tools - III	
	Toroidal Bend	
	Spinal Bend	
	Warp	
	Transform Tool	
	Warp Tool	
	Spine Tool	
	Stretch Tool	
	Bend Tool	
	Twist Tool	
11	Assembly Modeling	
	Assembly Modeling	
	Important Terms related to the Assembly Mode	
	Top-Down Approach	
	Bottom-up Approach	
	Placement Constraints	
	Package	
	Creating Top-down Assemblies	
	Creating Components in the Assembly Mode	
	Creating Bottom-up Assemblies	
	Inserting Components in an Assembly	
	Assembly Components	
	Displaying components in a separate window	
	Displaying components in the same window	
	3D Dragger	
	Applying Constraints	
	Status Area	
	Placement Tab	
	Move Tab	
	Packaging Components	
	Creating Simplified Representations	
	Redefining the components of an Assembly	

	Reordering Components	
	Supressing/resuming Components	
	Replacing Components	
	Assembling Repeated Copies of a component	
	Modifying the components of an assembly	
	Modyfying Dimensions of a feature of a component	
	Redefining a Feature of a component	
	Creating the Exploded state	
	References Tab Offset Tab	
	Explode line Tab	
	The Bill of Materials	
	Global Interference	
	Pairs Clearance	
12	Generating, Editing and Modifying the Drawing Views	
	The Drawing Mode	
	Generating Drawing Views	
	General View	
	Projection View	
	Detailed View	
	Auxillary View	
	Revolved Section View	
	Copy and Align View	
	3D Cross Section View	
	Editing the Drawing Views	
	Moving , Erasing and deleting the Drawing View	
	Adding new parts or Assemblies to the current Drawing	
	Modifying the Drawing Views	
	Changing the view Type	
	Changing the View Scale	
	Reorientating the Views	
	Modifying the Cross-sections	
	Modifying Boundaries of Views	
	Adding or removing the cross section Arrows	
	Modyfying the perspective Views	
	Modyfying Other parameters	
	Editing the Cross Section Hatching	
13	Dimensioning the Drawing Views	
	Show Model Annotations Dialog Box	
	Adding Notes to the Drawing	
	Adding Tolerances in the Drawing Views	
	Dimensional Tolerances	
	Geometric Tolerances	
	Editing the Geometric Tolerances	
	Adding Balloons to the Assembly Views	
	Adding Reference Datums to the Drawing Views	
	Modifying and Editing Dimensions	
	Modifying the Dimensions Using the Dimension Properties Dialog Box	
	Modifying the Drawing Items Using the Shortcut Menu	
	Cleaning Up the Dimensions	

14	Other Drawing Options	
	Sketching in the Drawing Mode	
	Modifying the Sketched Entities	
	User-Defined Drawing Formats	
	Retrieving the User-Defined Formats in the Drawings	
	Adding and Removing Sheets in the Drawing	
	Creating Tables in the Drawing Mode	
	Generating the BOM and Balloons in Drawings	
15	Surface Modeling	
	Creating Surfaces in Creo Parametric	
	Creating an Extruded Surface	
	Creating a Revolved Surface	
	Creating a Sweep Surface	
	Creating a Blended Surface	
	Creating a Swept Blend Surface	
	Creating a Helical Sweep Surface	
	Creating a Surface by Blending the Boundaries	
	Creating a Variable Section Sweep Surface Using the Sweep Tool	
	Creating Surfaces the Using the Style Environment of Creo Parametric	
	Style Dashboard	
	Surface Editing Tools	
	Mirroring the Surfaces	
	Merging the Surfaces	
	Trimming the Surfaces	
	Creating the Fill Surfaces	
	Creating the Intersect Curves	
	Creating the Offset Surfaces	
	Adding Thickness to a Surface	
	Converting a Surface into a Solid	
	Creating a Round at the Vertex of a Surface	
	Freestyle modelling environment	
	Freestyle Dashboard	
16	Working with Sheet Metal Components	
	Introduction to Sheet metal	
	Invoking The sheet metal Mode	
	Introduction to Sheet metal Walls	
	Creating the Planar Wall	
	Creating the Unattached Revolve Wall	
	Creating the Unattached Blend Wall	
	Creating the Unattached Offset Wall	
	Creating Reliefs in Sheet metal Components	
	Creating a Flat Wall Creating a Twist Wall	
	Creating an Extend Wall	
	Creating a Flange Wall	
	Creating the Bend Feature	
	Creating the Unbend Feature	
	Creating the Bend Back	
	Conversion to Sheet metal Part	
	Creating Cuts in the Sheet metal Components	